

Smart Metro Booking System

Project Planning Phase

2. Project Logic

1. Purpose

This document captures the core business and system logic driving the metro digital ticketing solution — specifically fare calculation and QR ticket generation/validation. These sit at the heart of the Fare Logic Script and QR Code Integration milestones from the project plan.

2. Fare Calculation Logic

Fare calculation kicks off once a passenger submits Source, Destination, and Passenger Type on the Service Catalog form. The flow below shows the step-by-step decision logic used to reach the final fare, including concession handling and multi-ticket bookings.

Figure 1 — Fare Calculation Logic Flow

2.1 Fare Logic Rules

Rule	Condition	Action
Base Fare	Always	Looked up from the Zone Fare Matrix using Source/Destination distance.
Student Concession	Passenger Type = Student	Apply the approved student discount % to the base fare.
Senior Citizen Concession	Passenger Type = Senior Citizen	Apply the approved senior citizen discount % to the base fare.
General Fare	Passenger Type = General	Apply the full standard fare, no discount.
Bulk Booking	Number of Tickets > 1	Multiply the resulting per-ticket fare by ticket count.
Invalid Route	Source = Destination	Reject the request and prompt the passenger to re-select stations.

3. QR Ticket Generation & Validation Logic

Once payment is confirmed, the system moves the QR ticket through a defined lifecycle — from generation, to active/unused status, to gate scans at entry and exit, and finally to closure and reconciliation. Any duplicate or expired scan attempt gets rejected.

Figure 2 — QR Ticket Generation & Validation State Logic

3.1 QR Lifecycle States

State	Trigger	Next State
Payment Confirmed	Successful payment gateway response	QR Generated
QR Generated	Unique, signed QR created	QR Active (Unused)
QR Active (Unused)	Passenger has ticket, not yet travelled	Scanned at Entry / Expired
Scanned at Entry	Valid QR scanned at AFC entry gate	In Transit
In Transit	Passenger inside metro network	Scanned at Exit
Scanned at Exit	Valid QR scanned at AFC exit gate	Trip Closed / Fare Reconciled
Invalid / Reused QR	Duplicate or expired scan attempt	Access Denied

4. Exception Handling Logic

- Payment failure — booking held in a pending state; passenger is prompted to retry or cancel; no QR is generated until payment succeeds.
- Duplicate QR scan — the gate rejects the scan, logs the attempt, and alerts the Station Manager for manual review.
- Expired QR — QR codes unused within the validity window are automatically invalidated and can't be scanned.
- Concession mismatch — if the passenger type can't be validated (e.g., missing ID), the system falls back to General fare and flags the transaction for review.
- Integration timeout — if the QR generation API doesn't respond within the defined threshold, the booking flow retries automatically before escalating to IT Admins.

5. Summary

Between the fare calculation logic and the QR lifecycle logic, every booking ends up with an accurate fare, a secure and traceable digital ticket, and a smooth entry-to-exit passenger experience — while Operations Teams and IT Admins retain the controls needed to handle exceptions safely.